

LIVE



Administrator: Command prompt

```
C:\# Azure Meetup  
@Casteddu
```



```
>Get-Date  
28 maggio 2026
```



```
>Get-Time  
14:30 - 15:30
```



```
>Get-Title  
"DNSSEC: dalla teoria  
all'implementazione con  
Azure DNS"
```

```
>$env:Speaker = "DAVIDE_RASOLI"
```





I nostri #sponsor





#Agenda

01

Sfatiamo i miti riguardanti DNSSEC

02

I componenti di DNSSEC

03

Come funziona DNSSEC

04

Da chi dipende la sicurezza del DNSSEC?

05

Implementazione di DNSSEC in Azure DNS



Daide Rasoli



VoIP & Cloud Specialist @ Com.tel SpA

Sono un ragazzo di 24 anni, da sempre appassionato di informatica, materia di indirizzo dell'istituto tecnico che ho frequentato. Ho poi iniziato a lavorare come sviluppatore di Web App, per poi orientarmi verso l'ambito più infrastrutturale / sistemistico.

[#azuremeetupcasteddu](#)

DNSSEC: sfatiamo i miti

DNSSEC protects recursive DNS

Falso

Ho già un WAF e un ottimo anti-spam, non ho bisogno di DNSSEC

Falso

DNSSEC protects DNS server
DNS is a target for hacker

Falso

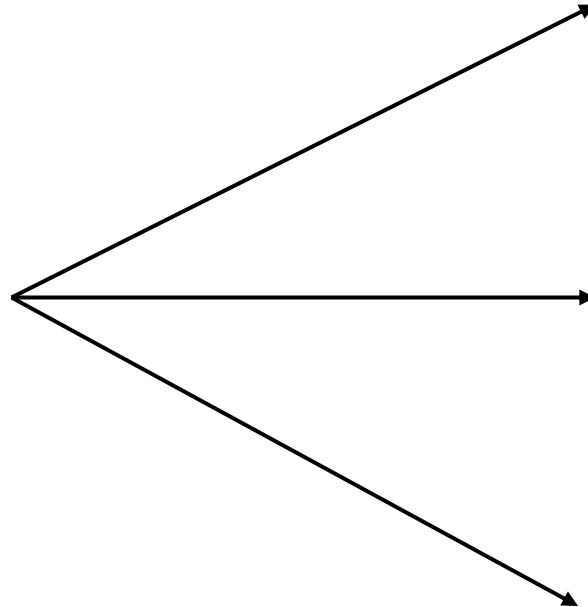
Configurare DNSSEC è molto
complesso e vale la pena solo alle
grandissime aziende

Falso

DNSSEC: utilizzo e vantaggi



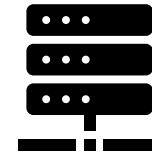
Consente ai client di validare le risposte DNS



«Protegge» da attacchi MITM

[Record DNS]

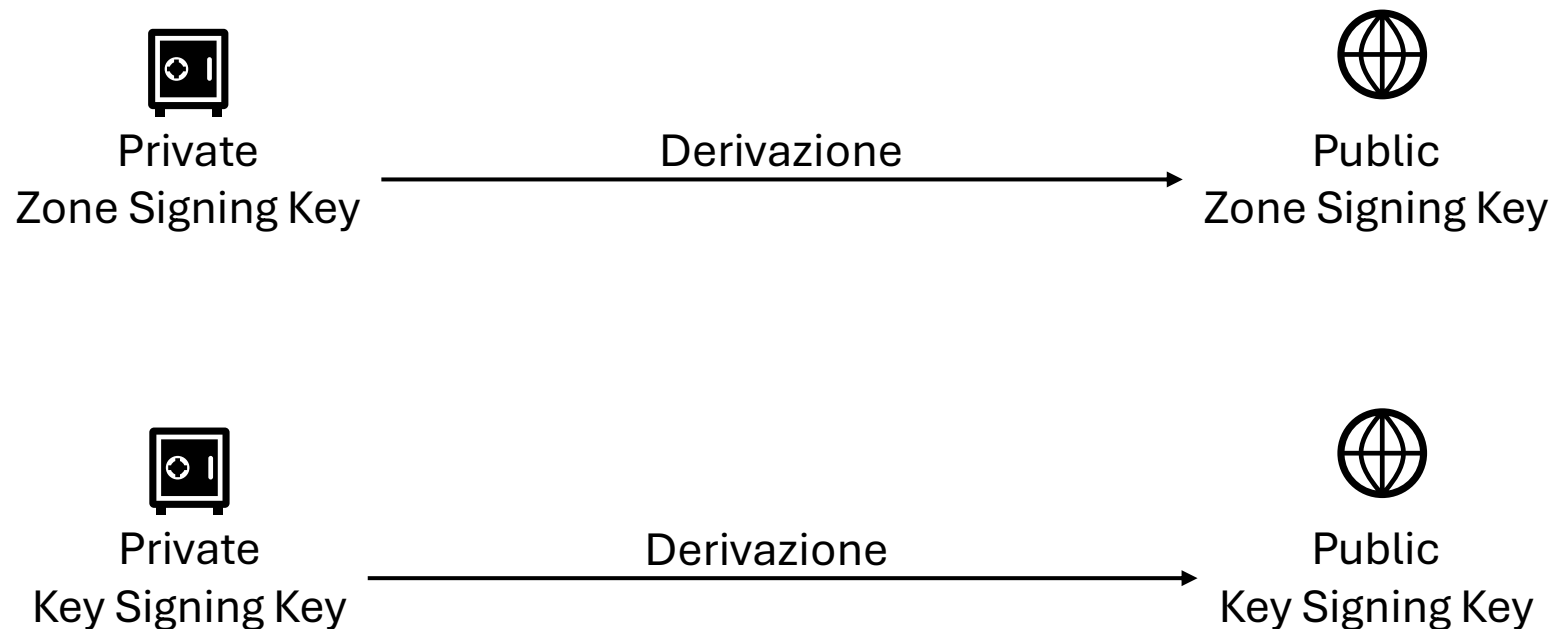
Esplicita l'inesistenza dei record



«Protegge» da DNS poisoning e cache spoofing

I componenti di DNSSEC

DNSSEC è un'estensione del protocollo DNS



I componenti di DNSSEC

Tipo record DNSKEY

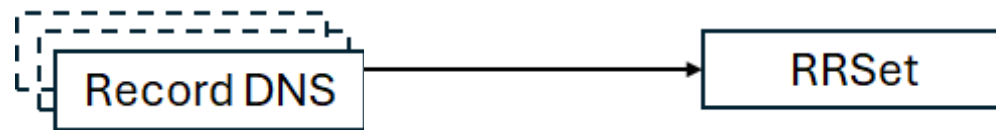
Host Label	Flags	Protocol	Algorithm	Public Key	Tags
example.com.	256	3	13	oJMRESz5E4gYzS/q6XDrvU1qMPYljCWzJaOau8XNEZeqCYKD5ar 0IRd8KqXXFJkqmVfRvMGPmM1x 8fGAa2XhSA==	34505
example.com.	257	3	13	mdsswUyr3DPW132mOi8V9xESWE8jTo0dxCjjnopKl+GqJxpVXck HAeF+KkxLbxILfDLUT0rAK9iUzy1L53eKGQ==	2371

Tipo record DS

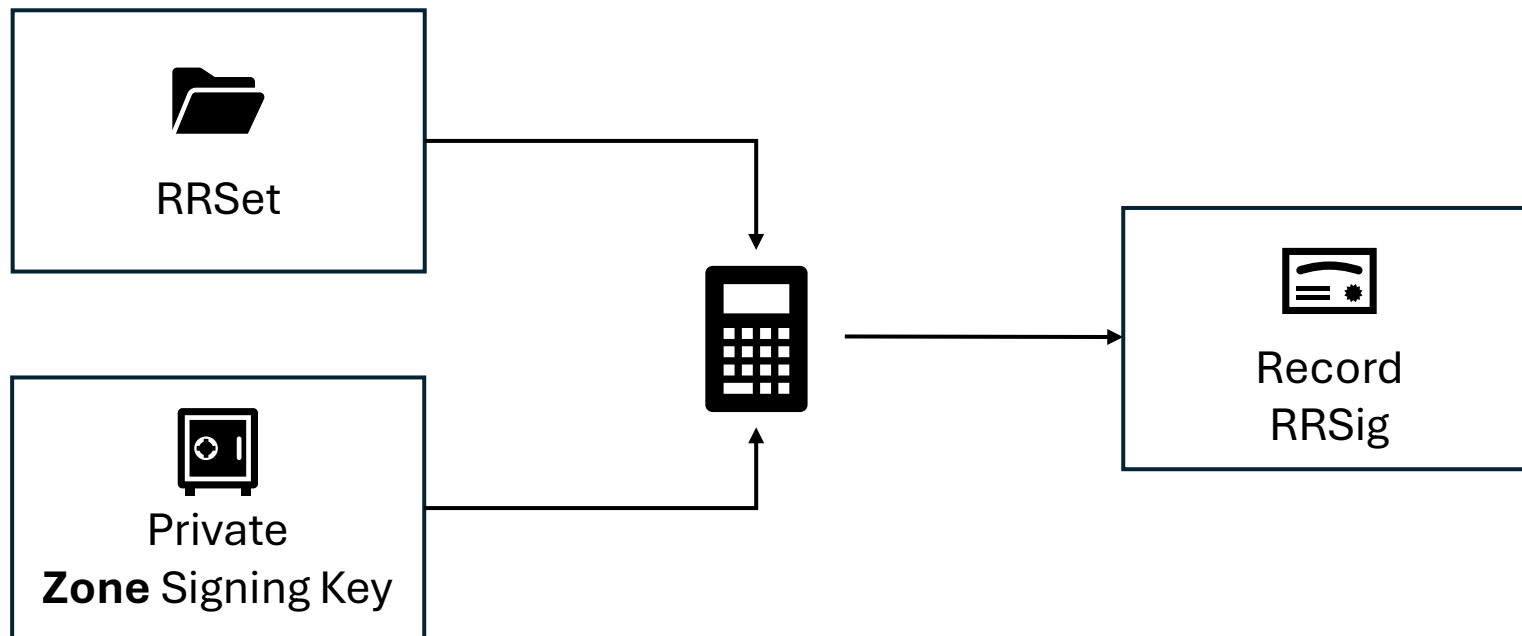
Host Label	Key Tag	Algorithm	Digest Type	Digest
example.com.	2371	13	2	c988ec423e3880eb8dd8a46fe06ca230ee 23f35b578d64e78b29c3e1c83d245a

Tipo record RRSig...

Come funziona DNSSEC

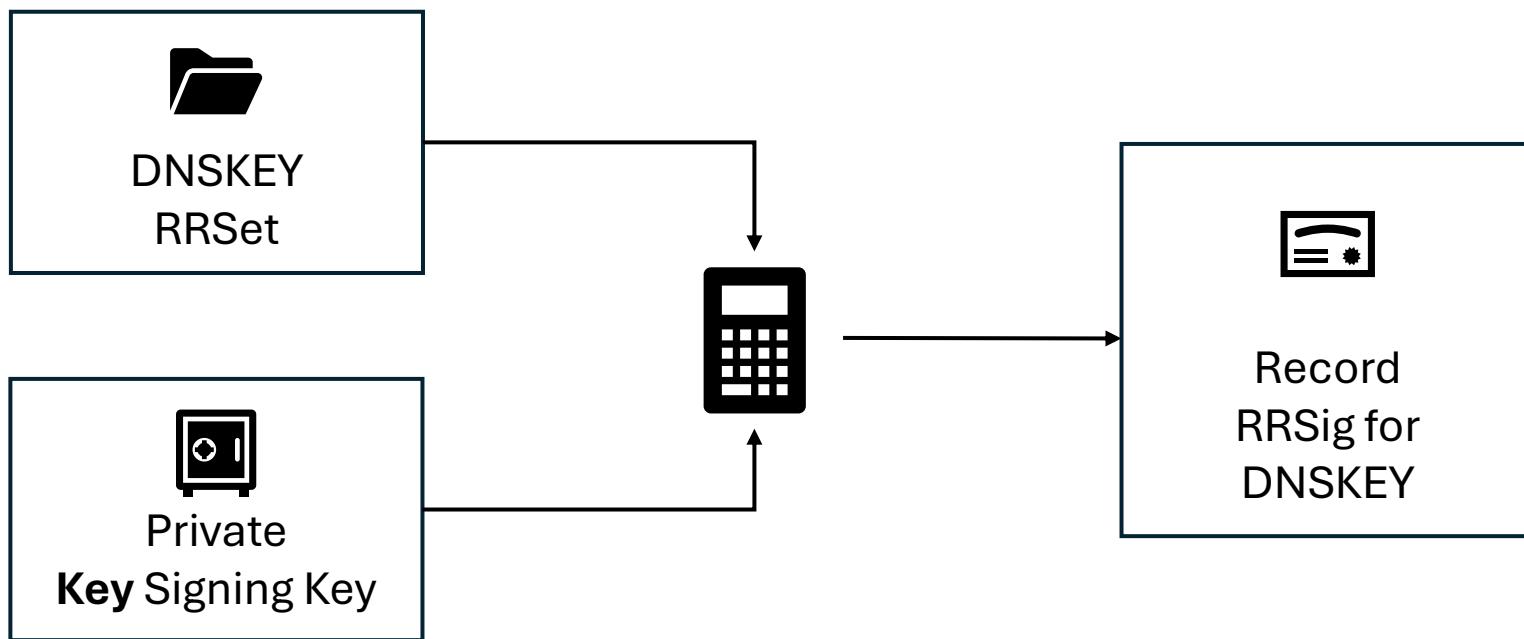


La firma degli RRSet «Standard»



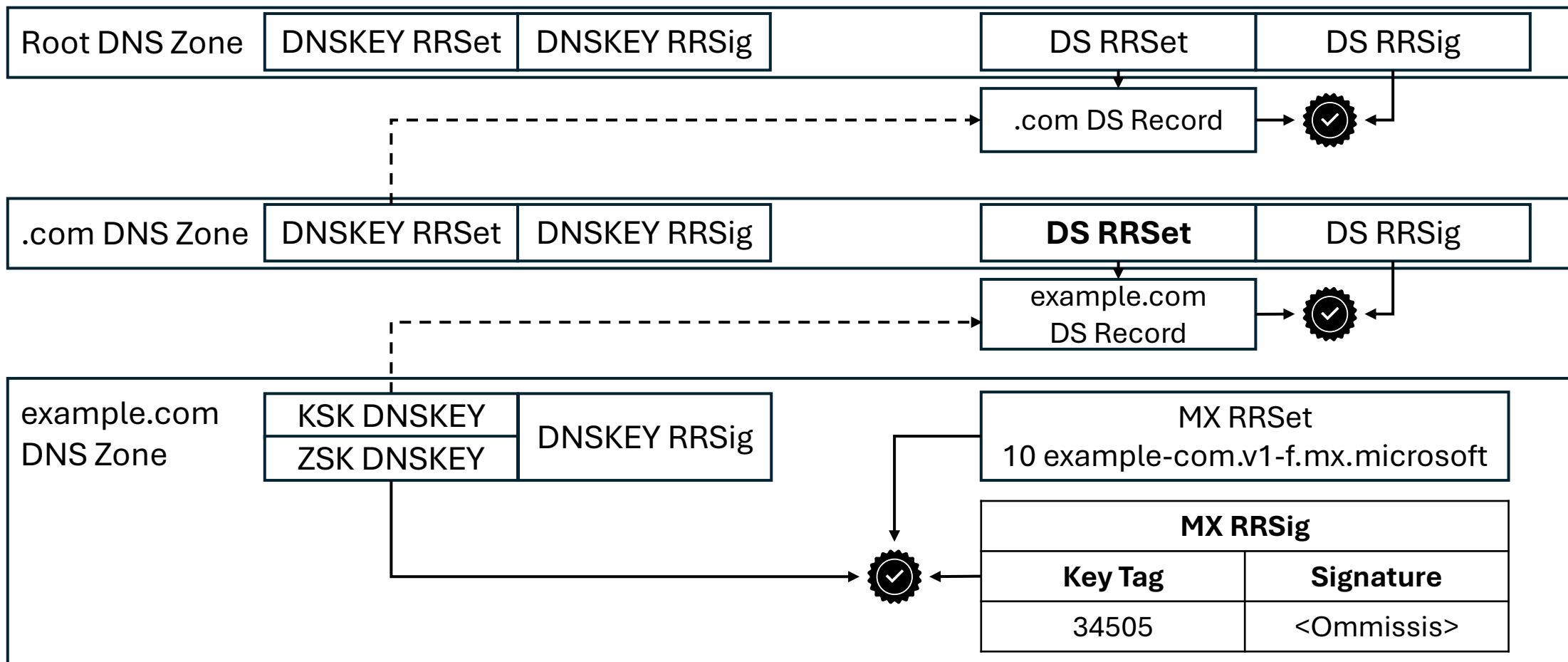
Come funziona DNSSEC

La firma del RRSset per i record DNSKEY



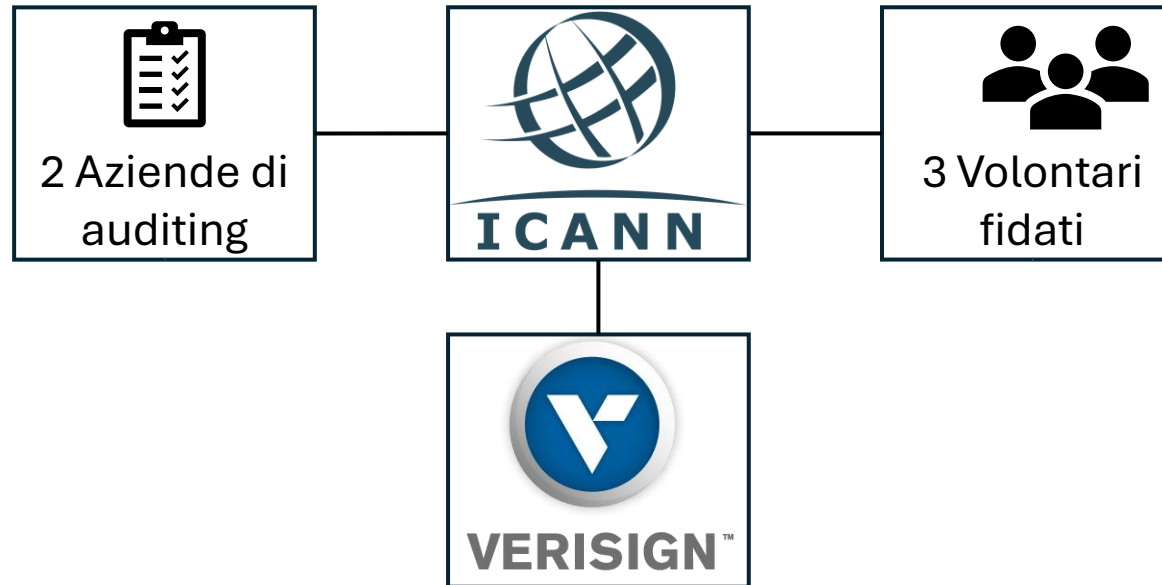
Come funziona DNSSEC

La catena di fiducia



Da chi dipende la sicurezza del DNSSEC?

L'origine della fiducia e la rotazione delle chiavi root



Data	Cerimonia	Agenda
2026-08-12	KSK Ceremony 62	Sign 2026Q4 ZSKs
2016-10-27	KSK Ceremony 27	Sign 2017Q1 ZSKs; Generate KSK-2017; Zeroize HSMs 1&2 (East)

#azuremeetupcasteddu

Configuriamo il DNSSEC in Azure

Step 0:

Dashboard > smus8.net

 **smus8.net** | DNSSEC ☆ ...

DNS zone

 Search

-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
- > Settings
- > DNS Management
 -  Recordsets
 -  **DNSSEC**
- > Monitoring
 -  Alerts
 -  Metrics
- > Automation
- > Help

DNSSEC stands for Domain Name System Security Extensions. It is a suite of extensions that adds security to your Azure DNS zone by enabling DNS responses to be validated. DNSSEC works by digitally signing DNS records to ensure they aren't forged or tampered with or during transit. When you enable DNSSEC, Azure DNS creates a key-signing key (KSK) and zone-signing key (ZSK) pair and then signs all the records in your zone using these keys. After enabling DNSSEC signing on your Azure DNS zone, you'll need to create a Delegation Signer (DS) record in the parent zone. This step is required to establish a chain of trust between your zone and the parent zone.

For more information:

[What is DNSSEC signing?](#)
[How to enable DNSSEC on my Azure DNS zone.](#)

Enable DNSSEC

☐

Domain Name: **smus8.net**

Detail: [more\(+\)](#) / [less\(-\)](#)

Analyzing DNSSEC problems for **smus8.net**

.	✓ Found 3 DNSKEY records for . ✓ DS=20326/SHA-256 verifies DNSKEY=20326/SEP ✓ DS=38696/SHA-256 verifies DNSKEY=38696/SEP ✓ Found 1 RRSIGs over DNSKEY RRset ✓ RRSIG=20326 and DNSKEY=20326/SEP verifies the DNSKEY RRset
net	✓ Found 1 DS records for net in the . zone ✓ DS=37331/SHA-256 has algorithm ECDSA256SHA256 ✓ Found 1 RRSIGs over DS RRset ✓ RRSIG=54393 and DNSKEY=54393 verifies the DS RRset ✓ Found 2 DNSKEY records for net ✓ DS=37331/SHA-256 verifies DNSKEY=37331/SEP ✓ Found 1 RRSIGs over DNSKEY RRset ✓ RRSIG=37331 and DNSKEY=37331/SEP verifies the DNSKEY RRset
smus8.net	✗ No DS records found for smus8.net in the net zone ✗ No DNSKEY records found ✓ ns1-02.azure-dns.com is authoritative for smus8.net ✗ No NSEC records in response ✗ No RRSIGs found
smus8.net	✓ ns3-02.azure-dns.org is authoritative for smus8.net ✗ No NSEC records in response ✗ No RRSIGs found
smus8.net	✓ ns4-02.azure-dns.info is authoritative for smus8.net ✗ No NSEC records in response ✗ No RRSIGs found
smus8.net	✓ ns2-02.azure-dns.net is authoritative for smus8.net ✗ No NSEC records in response ✗ No RRSIGs found

#azuremeetupcasteddu

Configuriamo il DNSSEC in Azure

Step 1:

Microsoft Azure

Dashboard > smus8.net

smus8.net | DNSSEC

DNS zone

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- DNS Management
 - Recordsets
 - DNSSEC**
- Monitoring
 - Alerts
 - Metrics
- Automation
- Help

DNSSEC stands for Domain Name System Security Extensions. It is a suite of extensions that adds security to your Azure DNS zone by enabling DNS responses to be validated. DNSSEC works by digitally signing DNS records to ensure they aren't forged or tampered with or during transit. When you enable DNSSEC, Azure DNS creates a key-signing key (KSK) and zone-signing key (ZSK) pair and then signs all the records in your zone using these keys. After enabling DNSSEC signing on your Azure DNS zone, you'll need to create a Delegation Signer (DS) record in the parent zone. This step is required to establish a chain of trust between your zone and the parent zone.

For more information:
[What is DNSSEC signing?](#)
[How to enable DNSSEC on my Azure DNS zone.](#)

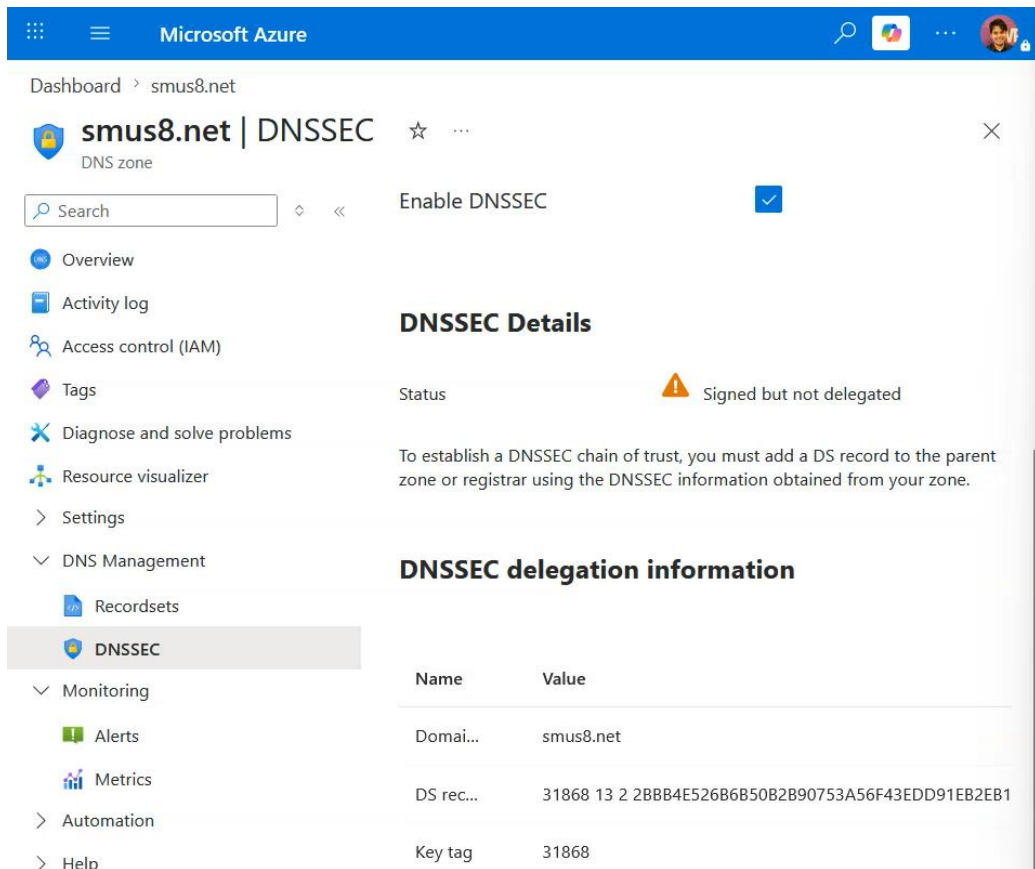
Enable DNSSEC ☐

Domain Name: smus8.net Detail: more(+) / less(-)	
Analyzing DNSSEC problems for smus8.net	
.	- Found 3 DNSKEY records for . - DS=20326/SHA-256 verifies DNSKEY=20326/SEP - DS=38696/SHA-256 verifies DNSKEY=38696/SEP - Found 1 RRSIGs over DNSKEY RRset - RRSIG=20326 and DNSKEY=20326/SEP verifies the DNSKEY RRset
net	- Found 1 DS records for net in the . zone - DS=37331/SHA-256 has algorithm ECDSA256SHA256 - Found 1 RRSIGs over DS RRset - RRSIG=54393 and DNSKEY=54393 verifies the DS RRset - Found 2 DNSKEY records for net - DS=37331/SHA-256 verifies DNSKEY=37331/SEP - Found 1 RRSIGs over DNSKEY RRset - RRSIG=37331 and DNSKEY=37331/SEP verifies the DNSKEY RRset
smus8.net	- No DS records found for smus8.net in the net zone - No DNSKEY records found - ns4-02.azure-dns.info is authoritative for smus8.net - No NSEC records in response - No RRSIGs found
smus8.net	- ns1-02.azure-dns.com is authoritative for smus8.net - No NSEC records in response - No RRSIGs found
smus8.net	- ns2-02.azure-dns.net is authoritative for smus8.net - No NSEC records in response - No RRSIGs found
smus8.net	- ns3-02.azure-dns.org is authoritative for smus8.net - No NSEC records in response - No RRSIGs found

#azuremeetupcasteddu

Configuriamo il DNSSEC in Azure

Step 2:



The screenshot shows the Microsoft Azure portal interface. The top navigation bar is blue with the Microsoft Azure logo. The main content area is titled "smus8.net | DNSSEC" and shows the "Enable DNSSEC" toggle switch, which is currently turned on (checked). Below this, the "DNSSEC Details" section displays the status as "Signed but not delegated" with a warning icon. A message explains that to establish a DNSSEC chain of trust, a DS record must be added to the parent zone or registrar. The "DNSSEC delegation information" table is also visible, showing the domain name, DS record, and key tag.

Name	Value
Domain	smus8.net
DS record	31868 13 2 2BBB4E526B6B50B2B90753A56F43EDD91EB2EB1
Key tag	31868



The screenshot shows the VeriSign Labs DNSSEC Debugger interface. The top navigation bar is blue with the VeriSign Labs logo. The main content area is titled "Domain Name:" and shows a text input field. Below the input field, there is a "Detail: more(+) / less(-)" link and a "Time: 2026-05-25 20:19:19 UTC" timestamp. The "DNSSEC Debugger" section is visible, along with a "Back to VeriSign Labs Tools" button. The "Advanced options" section is also present, with a "Legal Notices" link and a "Privacy" link.

Domain Name:

Detail: [more\(+\)](#) / [less\(-\)](#) Time: 2026-05-25 20:19:19 UTC

Enter a domain name to be tested.

[Back to VeriSign Labs Tools](#)

[Advanced options](#)

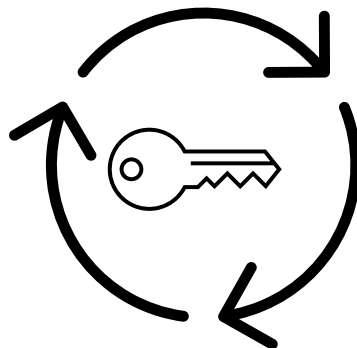
[Legal Notices](#) // [Privacy](#) // [Repository](#) // [Contact Us](#) // [Site Map](#)

© 2011-2026 VeriSign, Inc. All rights reserved.
VERISIGN, the VERISIGN logo, and other trademarks, service marks, and designs are registered or unregistered trademarks of VeriSign, Inc. and its subsidiaries in the United States and in foreign countries. All other trademarks are property of their respective owners.

#azuremeetupcasteddu

E poi?

Una volta implementato il DNSSEC, devo preoccuparmi di altro?



Rotazione delle chiavi ZSK e KSK

Link e documentazione per approfondire

- [Documentazione DNSSEC Microsoft Learn](#)
- [Spiegazione DNSSEC by CloudFlare](#)
- [Un ottimo sito per verificare e capire la catena DNSSEC](#)
- [Storico delle Root Signing Cerimonies](#)
- [Altre informazioni sulla Root Signing Ceremony](#)